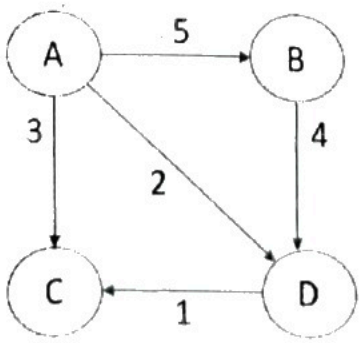
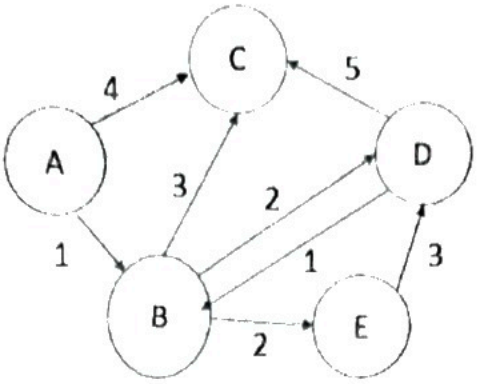
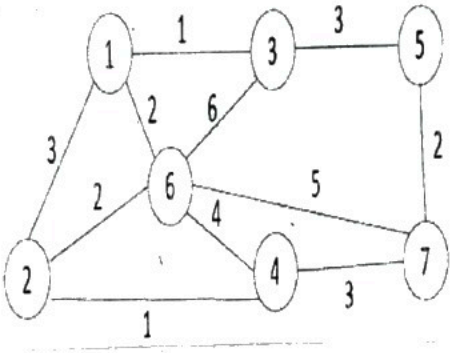
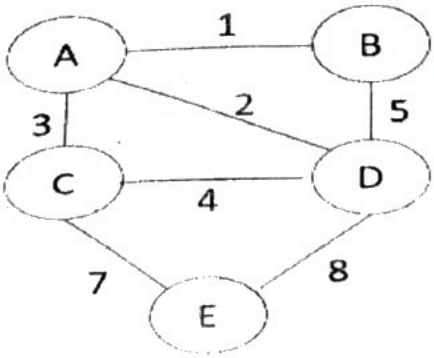
**VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE**
Matunga, Mumbai-400 019
**Autonomous Institute affiliated to University of Mumbai**

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|---|---|---|---|
| EXAMINATION | Mid Semester Examination (MSE) Sep 2024 | DATE OF EXAM | 24/09/24 |
| SEMESTER & PROGRAM | Sem-III, S.Y. B.Tech (Information Technology) | TIME | 11:15 AM to 12:45 PM |
| TIME ALLOWED | 1.30 HRS | MARKS | **40** |
| COURSE NAME – (CODE) | Design and Analysis of Algorithms – (R5IT2002T) | | |

Instructions 1. All questions carry indicated marks.
2. Figures to the right indicate full marks.

| Q.Nos | | Questions | Marks |
|---|---|---|---|
| 1 | a | Solve the following recurrence relations by using recursion tree method $T(n) = 2T\left(\frac{n}{2}\right) + n^3$ | [CO1] [5] |
| | b | State Master Theorem. Use the master method to solve the recurrence relation $T(n) = 2T\left(\frac{n}{4}\right) + \sqrt{n}$ | [CO1] [5] |
| 2 | a | Illustrate the heapsort algorithm and use it to sort the set {12,34,56,73,24,11,34,56,78,91}. | [CO1] [5] |
| | b | For the given graph implement the ALL-PAIRS-SHORTEST-PATH algorithm.   | [CO2] [7] |
| 3 | a | Write the Dijkstra's algorithm for finding the shortest path. Also, by using it find the shortest path from source **A** to all vertices present in the | [CO2] [8] |




| | | | |
|---|---|---|---|
| | b | Show step by step operation of Kruskal's algorithm on the following graph.   | [CO4] [5] |
| | c | Obtain Minimum Spanning Tree for the given graph by using Prim's algorithm.   | [CO4] [5] |

**********All the Best**********